



Alessandro Armando Vigliano

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Google Scholar

ORCID

Researchgate profile

RESEARCH FOCUS

Astrophysicist working on high-energy time-domain phenomena, with a particular focus on gamma-ray bursts, gamma-ray variability, and multi-instrument transient datasets. My research combines automated and reproducible analysis pipelines, statistical inference, and observational strategy design to study large heterogeneous datasets from space- and ground-based observatories. I am particularly interested in understanding how observational biases, variability properties, and selection effects influence the physical interpretation of transient populations, and in developing scalable methodologies for the analysis of X-ray and gamma-ray time-series data.

APPOINTMENTS

12/2024 – CURRENT

Postdoctoral Research Associate

Research within the Fermi, MAGIC, and CTAO collaborations.

Faculty of Physics, University of Trieste

Project: *"Spectral analysis of Gamma Ray Bursts and Solar Flares emitting in Gamma Rays"*

Responsible: Francesco Longo

EDUCATION

10/2021 – 10/2024

PhD in Mathematics and Physics

Faculty of Mathematical and Physical Sciences, University of Udine

Supervisor: Barbara De Lotto

Co-Supervisors: Francesco Longo, Stefano Ansoldi, Andrea Vacchi

Thesis: ***Novel strategies for the observations of fast gamma-ray transients***

2017 – 2021

Master's degree in Physics

Faculty of Physics, University of Trieste

Curriculum: Astrophysics and Cosmology

Thesis: ***Gamma Ray Burst polarization studies with AMEGO***

Supervisor: Stefano Ansoldi

Co-Supervisor: Francesco Longo

2014 – 2017

Bachelors's degree in Physics

Faculty of Physics, University of Trieste

Thesis: ***Estimation of the contribution of AGN host galaxies to reionization.***

Original title: *Stima del contributo di galassie ospitanti AGN alla reionizzazione.*

Supervisor: Pierluigi Monaco

Co-Supervisor: Fabio Fontanot

INTERNSHIPS

06/2020 – 09/2020

Megalib for simulation of polarized MeV photons: Master Internship

INFN, section of Trieste

02/07/2018 – 07/07/2018

Master Internship at Osservatorio Astrofisico e Astronomico di Asiago

Trieste University

01/09/2016 – 15/02/2017 **Satellite constellation system designing: Bachelor Internship**
PicoSaTs
<https://picosats.eu/>

SCHOOLS

08/10/2023 – 14/10/2023 **5th MAGIC Hardware School 2023**
MAGIC collaboration, La Palma, Canary Islands

12/02/2023 – 21/02/2023 **Winter School of Theoretical Physics and third COST Action CA18108 Training School "Gravity – Classical, Quantum and Phenomenology"**
COST Action CA 18108 QGMM, University of Wroclaw, Poland

03/09/2022 – 10/09/2022 **Quantum gravity phenomenology in the multi-messenger approach - COST CA18108 Second Training School**
COST Action CA18108 , Faculty of Physics, Belgrade (Serbia)

28/03/2022 – 08/04/2022 **Astrophysical sources of cosmic rays - ISAPP School Paris**
Institut Pascal, Université Paris Saclay

WORKSHOPS

03/04/2023 – 07/04/2023 **Workshop: Machine learning enabled searches for new physics in astrophysical data.**
IFPU, Trieste

09/01/2023 – 13/01/2023 **Workshop "High redshift Gamma-Ray-Bursts in the JWST era"**
Sexten Center for Astrophysics

18/07/2022 – 22/07/2022 **Workshop: Hands On the Extreme Universe with High Energy Gamma Rays**
Sexten Center for Astrophysics

23/05/2022 – 26/05/2022 **Gravitational Wave Open data Workshop 2022**
LIGO-Virgo-KAGRA, Trieste University

01/02/2022 – 03/02/2022 **International workshop on VHE Data Analysis using Open Source Packages**
University of Kashmir

TALKS, SEMINARS AND CONTRIBUTIONS

26/05/2026 – 28/05/2026 **Invited seminar**
Invited seminar: *Correlations in Gamma-Ray Bursts*
University of Palermo & INAF Palermo, Italy

24/03/2026 – 26/03/2026 **Theseus Conference 2026**
Contributed talk: *Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows*
Malaga, Spain

19/01/2026 – 23/01/2026 **Gamma-Ray Bursts: the Great Gig in the Sky**
Contributed talk: *Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows*
Sexten, Dolomites, Italy

17/11/2025 – 21/11/2025 **2nd LST+MAGIC F2F Science Meeting**
Contributed Talk: *The Soccer Ball Tessellation for the follow-up of Transient Events*
Granada, Spain

- 30/06/2025 – 04/07/2025 **Advances in Modeling High-Energy Astrophysical Sources: Insights from recent multimessenger discoveries**
Contributed Talk: *Insight on GRB physics from a novel data-driven method for systematic analysis of X-ray light-curves*
Sexten, Dolomites, Italy
- 24/03/2025 – 28/03/2025 **Celebrating 20 years of Swift Discoveries**
Presented a poster: *Insight on GRB physics from a novel data driven method for systematic analysis of X-ray light-curves*
Firenze, Italy
- 15/01/2024 – 17/01/2024 **1st VHEGAM meeting: LST proposals by the Italian community**
Contributed Talk: Alternative tiling strategies for transient events
LST collaboration, Bologna, Italy
- 26/07/2023 – 03/08/2023 **38th International Cosmic Ray Conference (ICRC2023)**
Presented a poster and proceeding on *Gamma Ray Burst localization with GECCO's BGO anticoincidence shields*
Institute for Cosmic Ray Research, University of Tokyo, Japan
- 19/06/2023 – 23/06/2023 **ASAPP2023 - Advances in Space AstroParticle Physics. Frontier technologies for particle measurements in space. International Conference**
Contributed Talk: *Gamma Ray Burst localization with GECCO's BGO anticoincidence shields*
Agenzia Spaziale Italiana (ASI), Istituto Nazionale di Fisica Nucleare (INFN), Perugia

SELECTED RESEARCH ACTIVITIES

- 01/2024 – ONGOING **Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows**
Collaborators: Francesco Longo, Zeljka Bosnjak
Paper
Development of a data-driven and model-independent framework for the automated analysis of Swift-XRT GRB afterglows. The pipeline enables homogeneous characterization of X-ray light curves for an order of magnitude more events than previous studies, allowing statistically robust investigations of GRB temporal and spectral correlations and of the impact of observational biases on population-level inference.
- 01/2024 – ONGOING **Development of the divergent mode observation for CTAO**
Collaborators: Francesco Longo & INFN Divergent Pointing team
Development and optimization of divergent pointing strategies for Imaging Atmospheric Cherenkov Telescopes, with applications to transient follow-up observations and wide-field extragalactic surveys for CTAO. Comparison with tessellation-based observational strategies and evaluation of survey efficiency and sky coverage.
- 11/2021 – 05/2025 **New tessellation strategies for multi-messenger observations of VHE fast transients with IACTs, PhD project**
Collaborators: Francesco Longo
PhD Thesis
Development of the "Soccer Ball Tessellation" strategy, a novel equal-area sky tessellation framework based on Goldberg polyhedra for the follow-up of poorly localized transient events such as GRBs and gravitational-wave counterparts. The method optimizes sky coverage while minimizing the number of telescope pointings, improving observational efficiency for narrow field-of-view instruments such as CTAO and other IACTs.

- 05/2021 – 08/2023 **GRB localization capability for GECCO.**
 Team: A. A. Vigliano, F. Longo, I. Martinez, E. A. Yates, A. Moiseev, A. Zoglauer.
 Proceeding, Poster
 Evaluation of GRB localization capabilities for the GECCO mission through simulations of the instrument BGO anti-coincidence shields. The study investigated the reconstruction of transient arrival directions from count-rate asymmetries and demonstrated localization accuracies of a few degrees on short timescales for GRB-like signals.
- 04/2020 – 05/2021 **Gamma Ray Burst polarization studies with AMEGO, Master's thesis.**
 Supervisor: Stefano Ansoldi
 Co-Supervisor: Francesco Longo
 Master Thesis
 Study of gamma-ray polarization signatures from GRBs using simulations for the AMEGO mission. Development of an improved instrument modulation model for polarization measurements in the transition regime between Compton scattering and pair production, with applications to the discrimination of competing GRB emission models.
- 02/2017 – 07/2017 **Estimation of the contribution of AGN host galaxies to Reionization; Undergraduate thesis.**
 Supervisor: Fabio Fontanot
 Co-Supervisor: Pierluigi Monaco
 Bachelor Thesis
 Study of the contribution of AGN host galaxies to the ionizing photon budget during the Epoch of Reionization, combining AGN luminosity functions, black hole-galaxy scaling relations, and star-forming galaxy models.

RESPONSIBILITIES

- 01/01/2026 – CURRENT **Joint Burst Advocate for the MAGIC & CTAO-LST1 Collaborations**
 La Palma, Canary islands
- 15/03/2024 – CURRENT **Member of the Board of Directors for the Astronomical Observatory "Helmut Ullric"**
 Società Astronomica Cortina, Col Drusciè, Cortina d'Ampezzo (Belluno), Italy
- 20/06/2024 – 31/12/2025 **Gravitational Wave Expert on Call for the MAGIC Collaboration**
 La Palma, Canary islands
- 15/01/2024 – 31/12/2025 **Burst Advocate for the CTAO-LST1 Collaboration**
 La Palma, Canary islands
- 15/01/2024 – 31/12/2025 **Burst Advocate for the MAGIC Collaboration**
 La Palma, Canary islands
- 13/01/2025 – 12/02/2025 **Data taking shift on MAGIC**
 Operator
 La Palma, Canary islands
- 29/01/2024 – 19/02/2024 **Data taking shift on CTAO-LST1**
 Deputy Shift Leader
 La Palma, Canary islands
- 04/08/2023 – 25/08/2023 **Data taking shift on CTAO-LST1**
 Operator
 La Palma, Canary islands

REFEREEING AND REVIEWING ACTIVITIES

- 2026 – CURRENT **Referee for Research in Astronomy and Astrophysics (RAA)**

2024 – CURRENT Referee for Physical Review D (APS)

2025 – CURRENT Internal reviewer for the MAGIC Collaboration

COLLABORATIONS

01/02/2025 – CURRENT **Member of the research team for the Crystal Eye proposal**
Evaluation of the localization capabilities of Gamma Ray Burst (GRB) signals

15/12/2024 – CURRENT **Member & co-proposer of the COST Action CA23130: Bridging high and low energies in search of quantum gravity (BridgeQG)**
Working Group 1: High-energy quantum gravity theory
Working Group 2: High-energy quantum gravity experiment

01/07/2023 – CURRENT **Member of the MAGIC Collaboration**
La Palma, Canary islands

01/11/2021 – CURRENT **Member of the CTAO-LST1 Collaboration**
La Palma, Canary islands

01/01/2020 – CURRENT **Member of the Astronomical Observatory "Helmut Ullric"**
Società Astronomica Cortina, Col Drusciè, Cortina d'Ampezzo (Belluno), Italy

01/08/2022 – 13/09/2023 **Member of the COST Action CA18108: QG-MM "Quantum gravity phenomenology in the multi-messenger approach"**
Working Group 3: Gamma rays

20/05/2021 – 30/09/2024 **Member of the research team for the Galactic Explorer with a Coded Aperture Mask Compton Telescope (GECCO).**
Evaluation of the localization capabilities of Gamma Ray Burst (GRB) signals using anti-coincidence shields (BGO)

TEACHING

2025 **Co-Supervisor for bachelor degree student thesis.**
"Analysis of AGN VHE emissions with the MAGIC telescopes"
University of Trieste, Trieste

2024 **Co-Supervisor for master degree student thesis.**
"Searching for high-z GRBs in Fermi-LAT data using ML techniques."
University of Trieste, Trieste

2023 – 2024 **External referee (INFN) for master degree student internships.**
5 students: MC simulations with Geant4 for medical physics
1 student: simulations of HE astrophysical signals, multiwavelength data analysis, CNN
University of Trieste, Trieste

02/10/2023 – 15/12/2023 **Teaching assistant for General Physics to the Mathematics curriculum**
University of Udine, Udine

27/02/2023 – 09/06/2023 **Teaching assistant for Analysis II to the Mathematics curriculum**
University of Udine, Udine

OUTREACH AND PUBLIC ENGAGEMENT

02/2026 **Lecture at high school**
Invited lecture on astrophysics and astronomy to high school students
Padova, Italy

09/2025 **German Italian Physics Exchange (GIPE)**

Presentation on gamma ray astronomy
University of Trieste, Italy

2022 – 2025 **Studiare Fisica a Trieste**

Presentation on gamma ray astronomy for incoming student orientation
University of Trieste, Italy

20/05/2025 **Pint of Science**

Public talk on gamma ray astrophysics: *Il Cielo nel Pallone*
Trieste, INFN & INAF, Italy

25/02/2025 – 26/02/2025 **2025 Italian Inter-regional Astronomy Olympics, judge and jury member.**
INAF, Italy

2021 – 2023 **Involved in the organization/staff of the joint INFN-INAF stand at the Trieste NEXT fair.**
University of Trieste, Italy

24/02/2022 – 26/02/2022 **2022 Italian Inter-regional Astronomy Olympics, judge and jury member.**
INAF, Italy

05/11/2021 – 13/11/2021 **International Remote Astronomy Olympics (IRAO 2021), Invigilator**
INAF, Italy

OTHER RESPONSIBILITIES

01/01/2021 – 29/09/2022 **Director of the competitive team and of relations with the national bodies CSEN and FIJLKAM. Representative of the athletes on the board of directors.**
Unione Judo Trieste

01/11/2016 – 28/11/2016 **First Sempai Seminar (Takemusu Aikido), assistant and organizer.**
Società Ginnastica Triestina

13/07/2015 – 27/07/2015 **Sailing Instructor**
Scuola Vela Golfo di Trieste, Muggia, Italy

SKILLS

LANGUAGES Italian – mother tongue
English – C1
• IELTS band 8 (23-24/07/2021)
Spanish – B1

DIGITAL SKILLS **Programming languages**
• **Python**
• Wolfram Mathematica
• Fortran (77, 90, 95)
• C
• Linux Bash
• IDL
• Arduino IDE
MEGAlib
AGI System Tool Kit (STK)
Git
SLURM

OTHER CERTIFICATIONS

BLSD - Basic Life Support and Defibrillation Basic Life Support with a Defibrillator, 23/10/2025

BLS - Basic Life Support Basic Life Support, 03/06/2025

Nautical license Nautical license, 04/07/2023, n° 953520

Scuba Diving PADI Open Water Diver, 24/08/2021, n° 2109EI8387

HOBBIES AND INTERESTS

Interests include martial arts, sailing, climbing, and music.